

Final examination
Introduction to Statistics
MA2140
Total marks 30
90 minutes

February 11, 2026

1. Consider a random sample of size n such that $X_1, \dots, X_n \sim_{i.i.d.} \text{truncated exponential}(\theta)$ with p.d.f. $f(x) = e^{-(x-\theta)}$, where $x > \theta$ and $\theta > 0$. Obtain the form of the maximum likelihood estimator (M.L.E.) of θ . Argue comprehensively.

(Marks: 7)

2. (a): Consider a normal population $X \sim N(\mu, \sigma^2)$. Consider a random sample $X_1, \dots, X_n$ from this population. Both μ and σ^2 are unknown. Comprehensively derive the form of the 95% confidence interval of μ .

(b): Suppose that when a signal having the true value μ is transmitted from location A, the value received at location B is normally distributed with mean μ and variance σ^2 . Here, σ^2 is unknown. Thus, if μ is sent, then the value received is $\mu + N$ where N , representing noise, is a normal distribution with mean 0 and variance σ^2 . To reduce error, suppose the same value is sent 9 times. If the successive values received are 5, 8.5, 12, 15, 7, 9, 7.5, 6.5, 10.5, construct a 95% confidence interval for μ . The upper $\alpha (= 0.025)$ point of a t-distribution with 8 d.f. is 2.3.

(Marks: 6+5)

3. A public health official claims that the mean home water use is 350 gallons a day. To verify this claim, a study of 20 randomly selected homes was followed up with the result that the average daily water uses of these 20 homes were as follows:

340	344	362	375
356	386	354	364
332	402	340	355
362	322	372	324
318	360	338	370

Do the data contradict the official's claim? Justify your answer using 10% level of significance. For your information, $t_{0.05,19} = 1.73$.

(Marks: 6)

4. Consider a two-sample problem. There are two populations: $X \sim N(\mu_1, \sigma_1^2)$ and $Y \sim N(\mu_2, \sigma_2^2)$. Here, σ_1^2 and σ_2^2 are known. You have two independent random samples of size n_1 and n_2 , drawn from these two populations: $(X_1, \dots, X_{n_1})$ and $(Y_1, \dots, Y_{n_2})$. Suppose you have to test for $H_0 : \mu_1 = \mu_2$ vs. $H_1 : \mu_1 \neq \mu_2$. Construct the critical region of this test with 5% level of significance. Provide all details.

(Marks: 6)

$$\sum_{i=1}^n x_i^2 + n\bar{x}^2 - 2\bar{x}\sum_{i=1}^n x_i$$

$$9.5 \quad \frac{(n-1)(\sum_{i=1}^n (x_i - \bar{x})^2)}{n} \quad \frac{(n-1)(\bar{x} - \mu)^2}{n}$$